Classification with ADALINE Network: Impact of Early Stopping

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1 Introduction

The ADALINE (Adaptive Linear Element) is a single-layer neural network that uses the Delta Rule for weight updates. Unlike the Perceptron, the error is calculated based on the continuous activation potential (u), allowing for the minimization of the Mean Squared Error (MSE) through gradient descent. This report analyzes the behavior of ADALINE using two optimization approaches: Stochastic Gradient Descent (SGD) and Batch Gradient Descent. Crucially, this analysis investigates a specific phenomenon observed during testing: how defining the stopping criterion by the absolute difference between the current and previous error ($|E_t - E_{t-1}| \leq \epsilon$) can trigger premature early stopping, preventing the model from achieving optimal test results.

2 Dataset 1: Linearly Separable Data

2.1 Error Evolution (MSE and Classification)

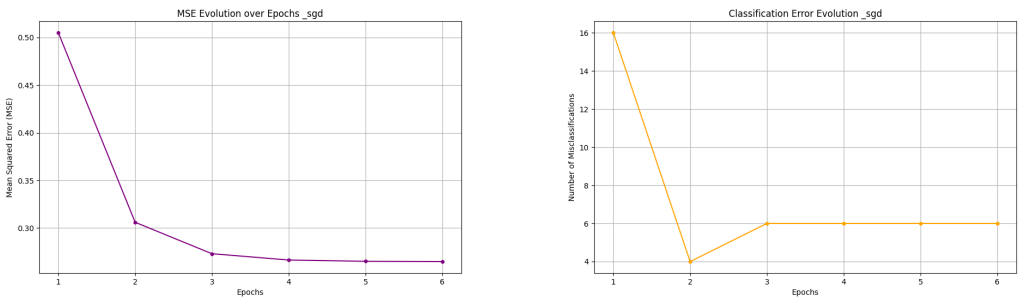

Figure 1: MSE and Classification Error Evolution (Stochastic).

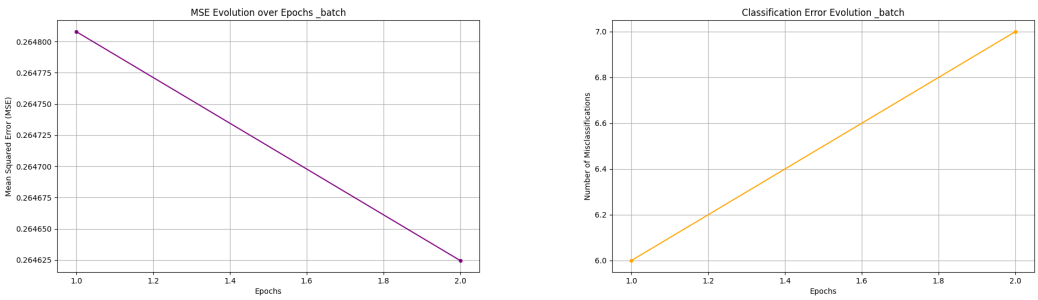

Figure 2: MSE and Classification Error Evolution (Batch).

2.2 Decision Boundaries (Train and Test)

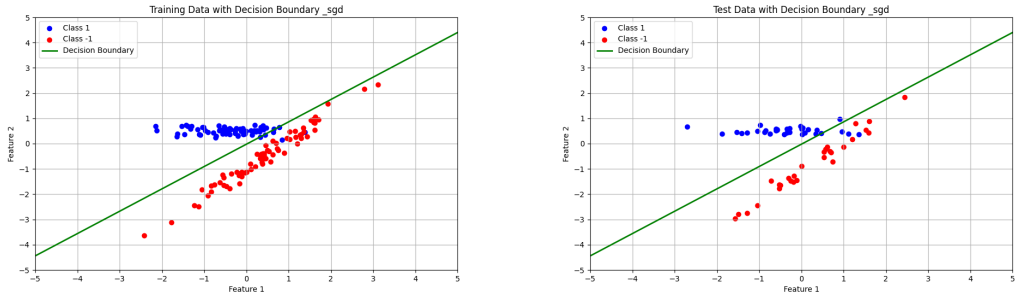

Figure 3: Decision Boundaries in Training and Testing (Stochastic).

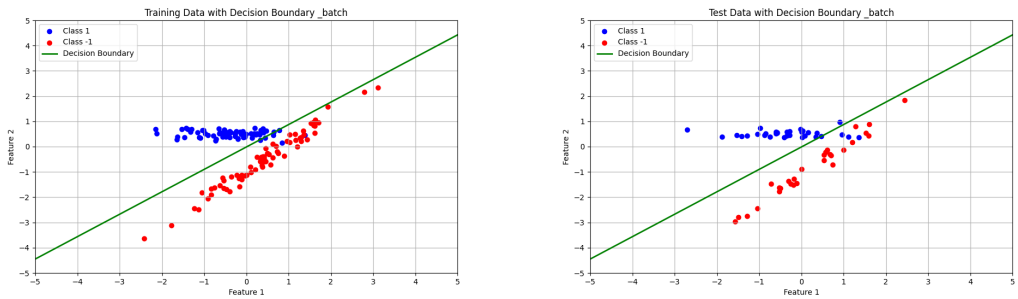

Figure 4: Decision Boundaries in Training and Testing (Batch).

2.3 Discussion of Results

Despite the data being linearly separable, the stopping criterion severely truncated the learning process.

Stochastic Gradient: Training stopped prematurely at epoch 6 due to the precision criterion. While the model achieved a high accuracy of 95.00% on the test set, it still recorded 6 misclassifications in the training set at the final epoch. This indicates that stopping based on the immediate error difference prevented the algorithm from finding the perfect linear separation.

Batch Gradient: Exhibited an even more drastic early stop at epoch 2. It achieved the same 95.00% test accuracy but with 7 training misclassifications. Because the Batch

method averages the gradients, the error descent is smoother; thus, the difference between epoch 1 and 2 was small enough to trigger the precision threshold almost immediately, halting convergence.

3 Dataset 2: Non-Linearly Separable Data

3.1 Error Evolution (MSE and Classification)

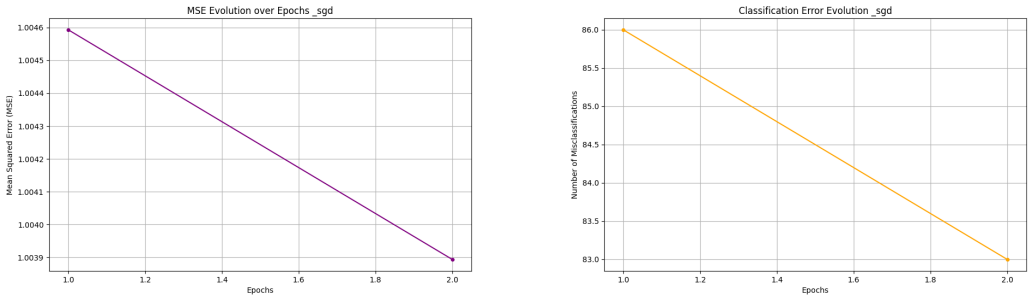

Figure 5: MSE and Classification Error Evolution (Stochastic).

3.2 Decision Boundaries (Train and Test)

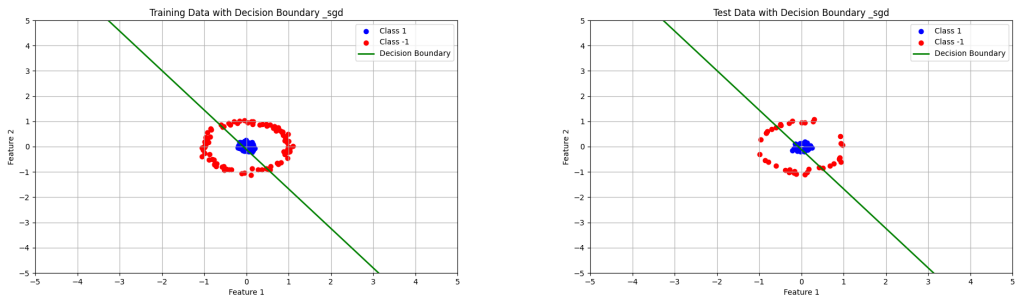

Figure 6: Decision Boundaries in Training and Testing with Noise (Stochastic).

3.3 Discussion of Results

In this dataset, the presence of noise creates overlapping regions between classes.

Stochastic Gradient: The noisy nature of this dataset caused the SGD to trigger the early stopping condition at just epoch 2. This yielded an accuracy of 61.33% on the test set with 83 misclassifications in training. The sample-by-sample update combined with noise caused immediate error fluctuation that mistakenly signaled convergence to the stopping function.

Batch Gradient: The batch approach accumulated the gradients, making it slightly more robust to local noise fluctuations and bypassing the premature stop of the initial epochs. It managed to train until epoch 40 before the precision criterion triggered. However, the final test accuracy was slightly lower (60.00%) with 87 misclassifications.

4 Dataset 3: High Dimensionality and Hyperparameters

4.1 Accuracy in Training and Test Sets

Table 1 presents the final metrics for each combination tested in Dataset 3 across 10 independent runs. Runs without a specific early stop epoch implies the model reached the maximum epoch limit.

Table 1: Mean Test Accuracies and Convergence Epochs by Configuration (Dataset 3)

Method	**Rate (η)**	**Precision (ϵ)**	**Stop Epoch**	**Mean Acc (%)**
Stochastic	0.01	0.1	3	87.30
Stochastic	0.01	0.0001	4	85.71
Stochastic	0.001	0.1	9	77.78
Stochastic	0.001	0.0001	35	87.30
Stochastic	0.0001	0.1	18	61.90
Stochastic	0.0001	0.0001	Max	79.37
Batch	0.01	0.1	Max	66.67
Batch	0.01	0.0001	Max	66.67
Batch	0.001	0.1	9	79.37
Batch	0.001	0.0001	35	85.71
Batch	0.0001	0.1	19	61.90
Batch	0.0001	0.0001	Max	79.37

4.2 Discussion of Results

In a high-dimensional scenario, the choice of hyperparameters clearly exposed the flaws of the relative error stopping criterion.

Impact of Precision (ϵ): When using a looser precision ($\epsilon = 0.1$), the model stopped too early across both methods. For instance, with $\eta = 0.001$, SGD stopped at epoch 9 yielding only 77.78% accuracy. By tightening the precision to $\epsilon = 0.0001$, the model was forced to train until epoch 35, which significantly improved the test accuracy to 87.30%.

Method Comparison: Overall, SGD achieved the highest peak accuracy (87.30%) compared to Batch (85.71%). Batch training struggled particularly with larger learning rates ($\eta = 0.01$), stalling at 66.67% accuracy. This reinforces that while Batch is more stable, SGD's ability to navigate complex error surfaces allows it to find better minima in high-dimensional spaces, provided the stopping criterion does not halt it prematurely.

5 Conclusion

This activity demonstrated the behavior of the ADALINE network under different optimization techniques. The most critical finding of this report is that utilizing the absolute difference between consecutive epoch errors ($|E_t - E_{t-1}|$) as a stopping criterion is a flawed approach. As observed across all three datasets, the learning curve can flatten temporarily or descend slowly, triggering a stop long before the global minimum is reached.

While Stochastic Gradient Descent generally achieved better peak accuracies than Batch Gradient Descent in high-dimensional spaces, both methods were heavily bottlenecked by premature stopping. Future implementations should consider alternative stopping criteria, such as evaluating the error on a separate validation set or using a "patience" threshold that waits for multiple epochs of non-improvement before halting the training.